

Quantum singular value transformation and its applications

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PERSONAL DATA

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RESEARCH INTERESTS

My research spans several areas at the intersection of condensed matter physics and quantum information science. I specialize in the physics of topological systems and their applications to quantum technologies, particularly in the realm of topological quantum computation and quantum sensing. I have made significant contributions to broader areas within quantum information science, including quantum algorithms and implications of PT-symmetric quantum mechanics.

Keywords: topological insulators and superconductors, bulk-boundary correspondence, Majorana qubits, quantum algorithms, quantum sensing, PT-symmetric quantum mechanics.

EMPLOYMENT HISTORY

08/2025 - present	Concordia University , Canada Assistant Professor
08/2022 - 05/2025	University of Sydney , Australia Postdoctoral Research Fellow
	University of Calgary , Canada
09/2020 - 07/2022	Killam Postdoctoral Fellow
03/2019 - 08/2020	Postdoctoral associate
01/2022 - 04/2022	Sessional Instructor (part-time)

EDUCATION

09/2013 - 02/2019	Dartmouth College , USA Ph.D. in Physics (March 2nd, 2019) Supervisor: Prof. Lorenza Viola Dissertation: Boundary physics and bulk-boundary correspondence in topological phases of matter
08/2009 - 05/2013	Indian Institute of Technology , Bombay, India B.Tech. with Honors in Engineering Physics (August 2012)

Many important quantum computational tasks require implementing a function g of a Hermitian matrix H on a given state $|\psi\rangle$.

- Hamiltonian simulation: Given a d -sparse Hermitian matrix H , an initial state $|\psi\rangle$, and a time t , generate the state $e^{-iHt}|\psi\rangle$ up to a given error ε .
- System of linear equations: Given a d -sparse matrix A and an initial state $|\mathbf{b}\rangle$, generate the normalized state $|\mathbf{x}\rangle \propto A^{-1}|\mathbf{b}\rangle$ up to a given error ε . This problem is equivalent to generating a state proportional to $H^{-1}|0\rangle|\mathbf{b}\rangle$, where $H = |0\rangle\langle 1| \otimes A + |1\rangle\langle 0| \otimes A^\dagger$.
- Other applications: Gibbs sampling, Differential equation solvers, SDP solvers, ...

Problem statement (state generation setting)

Given a function $g : [-1, 1] \rightarrow \mathbb{C}$ with $\|g\|_\infty \leq 1$, a d -sparse n -qubit Hermitian matrix H with $\|H\| \leq 1$, and a circuit to prepare an n -qubit state $|\psi\rangle$, generate a normalized state proportional to $g(H)|\psi\rangle$ up to error ε .

Binary oracle for f

A black-box function f is specified by

$$O_f |z\rangle |0\rangle = |z\rangle |f(z)\rangle.$$

If f is efficiently computable classically, then O_f can be implemented efficiently.

Sparse oracles for matrices

A sparse matrix A can be specified by two functions:

$$\begin{aligned} A_{\text{idx}} : (i, k) &\mapsto r_{ik}, & r_{ik} &\text{ is the } k\text{th non-zero entry in row } i \\ A_{\text{val}} : (i, j) &\mapsto [A_{ij}]_b & [A_{ij}]_b &\text{ is the } b\text{-bit binary representation of } A_{ij}. \end{aligned}$$

Oracles for A_{idx} and A_{val} specify the matrix A .

However, processing sparse oracles is quite challenging and unintuitive.

Definition: Block encoding

For a $2^n \times 2^n$ matrix A and $\alpha, \epsilon \in \mathbb{R}^+$, we say that an $(n+a)$ -qubit unitary operator U_A is a (α, a, ϵ) -block encoding of A if

$$|A - \alpha \langle 0^a | U_A | 0^a \rangle| \leq \epsilon.$$

- Equivalently,

$$U_A \approx \begin{bmatrix} A/\alpha & * \\ * & * \end{bmatrix} \quad \text{or} \quad \boxed{A/\alpha} \approx \begin{array}{c} \text{---} \\ \boxed{U_A} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ |0^a\rangle \\ \text{---} \end{array}.$$

Note: The time arrow points to the left (Kitaev notation).

- Block encoding is a convenient way of specifying a matrix input to a quantum algorithm is via block encoding¹.
- If A is sparse and its entries are efficiently computable, then a description of the circuit implementing U_A can be computed efficiently classically.
- Block encodings have many interesting and useful properties. For example, given U_A and U_B , we can efficiently construct U_{A+B} and U_{AB} .
- Oblivious amplitude amplification: The scaling factor can be improved to $\mathbf{O}(1)$ at cost $\mathbf{O}(1/\alpha)$.

¹G. H. Low and I. L. Chuang, Quantum 3, 163 (2019).

Lemma 48 (Block-encoding of sparse-access matrices). *Let $A \in \mathbb{C}^{2^w \times 2^w}$ be a matrix that is s_r -row-sparse and s_c -column-sparse, and each element of A has absolute value at most 1. Suppose that we have access to the following sparse-access oracles acting on two $(w + 1)$ qubit registers*

$$\begin{aligned} O_r: |i\rangle|k\rangle &\rightarrow |i\rangle|r_{ik}\rangle & \forall i \in [2^w] - 1, k \in [s_r], \text{ and} \\ O_c: |\ell\rangle|j\rangle &\rightarrow |c_{\ell j}\rangle|j\rangle & \forall \ell \in [s_c], j \in [2^w] - 1, \text{ where} \end{aligned}$$

r_{ij} is the index for the j -th non-zero entry of the i -th row of A , or if there are less than i non-zero entries, then it is $j + 2^w$, and similarly c_{ij} is the index for the i -th non-zero entry of the j -th column of A , or if there are less than j non-zero entries, then it is $i + 2^w$. Additionally assume that we have access to an oracle O_A that returns the entries of A in a binary description

$$O_A: |i\rangle|j\rangle|0\rangle^{\otimes b} \rightarrow |i\rangle|j\rangle|a_{ij}\rangle \quad \forall i, j \in [2^w] - 1, \text{ where}$$

a_{ij} is a b -bit binary description²² of the ij -matrix element of A . Then we can implement a $(\sqrt{s_r s_c}, w + 3, \varepsilon)$ -block-encoding of A with a single use of O_r, O_c , two uses of O_A and additionally using $\mathcal{O}(w + \log^{2.5}(\frac{s_r s_c}{\varepsilon}))$ one and two qubit gates while using $\mathcal{O}(b, \log^{2.5}(\frac{s_r s_c}{\varepsilon}))$ ancilla qubits.

Lemma 12 (Block encoding amplification [60, Theorem 2]). *Let C be a matrix such that C/α is block encoded by O_C with some normalization factor α . Then given any $\alpha_C \geq \|C\|$, the operator*

$$\frac{C}{2\alpha_C} \tag{123}$$

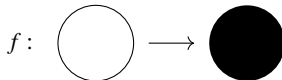
can be block encoded with accuracy ε using

$$\mathcal{O}\left(\frac{\alpha}{\alpha_C} \log\left(\frac{1}{\varepsilon}\right)\right) \tag{124}$$

queries to the controlled- O_C and its inverse.

Function of a black-box unitary task

Given a function $f: S^1 \rightarrow \mathbb{C}$ with $\|f\|_\infty \leq 1$ and an n -qubit unitary U , construct a (α, a, ε) -block encoding of $f(U)$ with $\alpha \in \mathbf{O}(1)$.



Laurent polynomials

For most of the talk, we will focus on $f(z) = \sum_{l=-d}^d \beta_l z^l$. Laurent polynomials are great for approximating functions on the unit circles, i.e. convergence with d is fast.

Function of a black-box unitary task

Given a function $f : S^1 \rightarrow \mathbb{C}$ with $\|f\|_\infty \leq 1$ and an n -qubit unitary U , construct a (α, a, ε) -block encoding of $f(U)$ with $\alpha \in \mathbf{O}(1)$.

- That is, we want to construct an approximation to

$$\mathcal{U} = \begin{bmatrix} f(U)/\alpha & * \\ * & * \end{bmatrix} \quad \text{or} \quad \boxed{f(U)/\alpha} \approx \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \boxed{\mathcal{U}} \\ \boxed{\mathcal{U}} \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} \\ \langle 0^a | \end{array} \begin{array}{c} \\ | 0^a \rangle \end{array}$$

with α bounded from above by a constant independent of f .

- The figure of merit is query complexity, which is the number of times controlled U appears in the block encoding of $f(U)$.

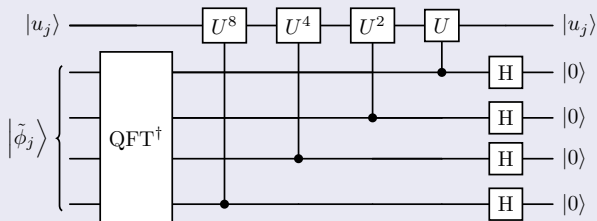
Quantum fourier transform

$$|\phi_k\rangle \left\{ \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right\} \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \left. \vphantom{\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array}} \right\} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \left. \vphantom{\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array}} \right\} |k\rangle \Rightarrow |k\rangle \left\{ \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right\} \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \left. \vphantom{\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array}} \right\} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \left. \vphantom{\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array}} \right\} |\phi_k\rangle$$

where $|\phi_k\rangle := \frac{1}{\sqrt{2^n}} \sum_j e^{2\pi i k j / 2^n} |j\rangle$

Quantum phase estimation

$$U|u\rangle = e^{2\pi i \theta} |u\rangle.$$



The candidates for eigenvalues are $\{z_k = \exp(2\pi i k/2^m), k = 0, \dots, 2^m - 1\}$, so the worst case error is at least $1/\varepsilon \in \mathbf{O}(2^m)$. Also, the circuit uses $2^m - 1$ queries. So the query complexity is $\mathbf{O}(1/\varepsilon)$, not $\text{polylog}(1/\varepsilon)$.

Furthermore, the phase register is in a superposition of good and bad guesses, which is additional error. Suppose $U|u\rangle = z|u\rangle$. Then

$$\begin{aligned} |u\rangle \frac{1}{\sqrt{2^m}} \sum_{l=0}^{2^m-1} |l\rangle &\mapsto |u\rangle \frac{1}{\sqrt{2^m}} \sum_{l=0}^{2^m-1} z^l |l\rangle \\ &\mapsto |u\rangle \frac{1}{\sqrt{2^m}} \sum_{l=0}^{2^m-1} z^l \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} z_k^{-l} |k\rangle \\ &= |u\rangle \sum_{k=0}^{2^m-1} \left(\frac{1}{2^m} \sum_{l=0}^{2^m-1} (z/z_k)^l \right) |k\rangle \\ &= |u\rangle \sum_{k=0}^{2^m-1} \left(\frac{1}{2^m} \frac{1 - (z/z_k)^{2^m}}{1 - (z/z_k)} \right) |k\rangle. \end{aligned}$$

This source of error can be tackled by repetition and majority voting.

Binary oracle for f

A black-box function f is specified by

$$O_f |z\rangle |0\rangle = |z\rangle |f(z)\rangle.$$

If f is efficiently computable classically, then O_f can be implemented efficiently.

Diagonal encoding of f

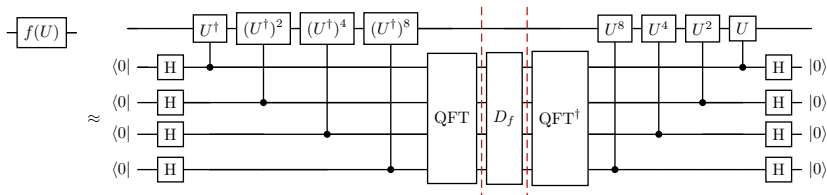
O_f can be used to construct a block encoding of D_f , where

$$[D_f]_{kk} = f(z_k), \quad \{z_k = \exp(2\pi i k / 2^m), k = 0, \dots, 2^m - 1\}.$$

$$\boxed{D_f} = \langle 0^a | \boxed{U_f} | 0^a \rangle$$

Quantum phase estimation (QPE) approach

QPE can be used to block encode $f(U)^2$.



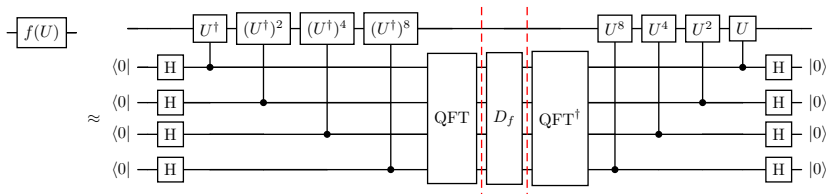
$$f(U) \approx \text{QPE}^\dagger * D_f * \text{QPE}.$$

$$f(\theta_j) \approx \langle \tilde{\theta}_j | \left(\sum_k f(z_k) |k\rangle \langle k| \right) | \tilde{\theta}_j \rangle.$$

²A. W. Harrow, A. Hassidim, and S. Lloyd, Phys. Rev. Lett. 103, 150502 (2009). [↗](#) [↖](#) [↻](#) [🔍](#)

Quantum phase estimation (QPE) approach

QPE can be used to block encode $f(U)^2$.



Pros

- Constant scaling factor for block encoding of $f(U)$.
- Works for black-box function f . No need to access a (Laurent) polynomial approximation.
- Only one call to U_f .

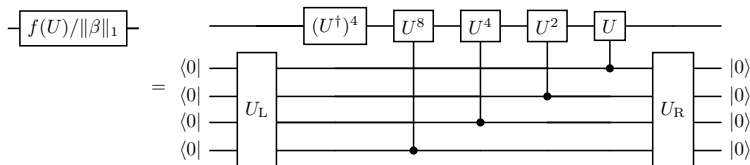
Cons

- $\log(d) + 2$ ancilla qubits (this is a choice we made).
- With $4d - 1$ uses of U , the precision achieved is $1/\epsilon \in \text{poly}(d)$.

²A. W. Harrow, A. Hassidim, and S. Lloyd, Phys. Rev. Lett. 103, 150502 (2009).

Linear combination of unitaries (LCU) approach

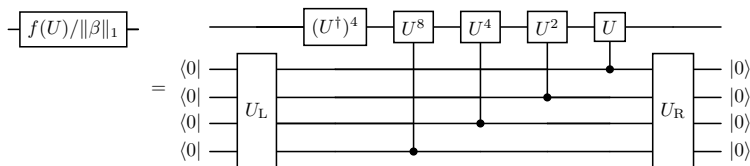
Let $f(z) = \sum_{l=-d}^d \beta_l z^l$. Then



$$\frac{1}{\|\beta\|_1} \sum_l \beta_l U^l = \underbrace{\frac{1}{\sqrt{\|\beta\|_1}} \left(\sum_k \sqrt{\beta_{k-d}} \langle k|_a \right)}_{\langle 0|U_L} * \underbrace{\left(\sum_j |j\rangle_a \langle j| \otimes U^{j-d} \right)}_{\mathcal{W}_U} * \underbrace{\frac{1}{\sqrt{\|\beta\|_1}} \left(\sum_m \sqrt{\beta_{m-d}} |m\rangle_a \right)}_{U_R|0\rangle}$$

Linear combination of unitaries (LCU) approach

Let $f(z) = \sum_{l=-d}^d \beta_l z^l$. Then



Pros

- A Laurent polynomial of degree d can be implemented exactly with $O(d)$ uses of U . If f is analytic in a neighborhood of S^1 , then the error decreases exponentially in d .

Cons

- Scaling factor $\|\beta\|_1 \in \mathbf{O}(\sqrt{d})$.
- Need access to the coefficients β_j .
- $\log(d)$ ancilla qubits.

We can express the signal operator in the σ_x basis by applying a Hadamard gate to the both sides of the ancilla qubit:

$$A_{\sigma_x} := \begin{bmatrix} \cos(\mathcal{H}) & i \sin(\mathcal{H}) \\ i \sin(\mathcal{H}) & \cos(\mathcal{H}) \end{bmatrix} = (H \otimes I) A_{\sigma_z} (H \otimes I)$$

This shows that, at a high level, both of these formalisms are conceptually identical although there are distinctions between the forms of the polynomials that can be constructed in both bases. The results of QSP are often stated in the A_{σ_x} formalism as:

Theorem 1 (Quantum Signal Processing in σ_x Basis). $\forall d \in \mathbb{N}$ and $\vec{\phi} \in \mathbb{R}^{d+1}$:

$$\left(\prod_{j=1}^d R_z(\phi_j) A_{\sigma_x} \right) R_z(\phi_0) = \begin{bmatrix} P(\cos(\mathcal{H})) & iQ(\cos(\mathcal{H})) \sin(\mathcal{H}) \\ iQ^*(\cos(\mathcal{H})) \sin(\mathcal{H}) & P^*(\cos(\mathcal{H})) \end{bmatrix} \quad (2)$$

$$\Longleftrightarrow$$

- (I) P and Q are polynomials s.t. $\deg(P) \leq d$ and $\deg(Q) \leq d - 1$.
- (II) P has parity $d \bmod 2$ and Q has parity $(d - 1) \bmod 2$.
- (III) $\forall x \in [-1, 1], |P(x)|^2 + (1 - x^2)|Q(x)|^2 = 1$

Theorem 2 (Quantum Signal Processing in σ_z Basis). $\forall P \in \mathbb{R}[x]$, with $\deg(P) = d$, $\exists \vec{\phi} \in \mathbb{R}^{d+1}$ s.t.:

$$\forall x \in \mathbb{T}, |P(x)|^2 \leq 1 \wedge \text{Parity}(P) = d \bmod 2 \quad (3)$$

$$\Longleftrightarrow$$

$$\begin{bmatrix} P(e^{i\mathcal{H}}) & \cdot \\ \cdot & \cdot \end{bmatrix} = \left(\prod_{j=1}^d R_x(\phi_j) A_{\sigma_z} \right) R_x(\phi_0) \quad (4)$$

Here $\mathbb{T} = \{x \in \mathbb{C} : |x| = 1\}$ is again the complex unit circle.

The above formalism indeed mitigates some previously discussed limitations. However, the requirement that the polynomial P must be real and possess definite parity necessitates the combination of four separate instances of QSP using Linear Combination of Unitaries (LCU) to construct an arbitrary complex polynomial with indefinite parity. In the following section, we address these constraints with a novel approach. We propose a more generalized formulation of Quantum Signal Processing by allowing our signal processing operators to be arbitrary $\text{SU}(2)$ rotations, effectively lifting the "realness" condition on the polynomials. Furthermore, we eliminate the parity restriction by simplifying our signal operator. Combined together, our reformulation successfully eliminates all practical restrictions on P , which we will elaborate on and demonstrate in the next section.

III. GENERALIZED QUANTUM SIGNAL PROCESSING

Our generalized quantum signal processing approach seeks to remove the limitations inherent in traditional QSP, thereby providing a more powerful mechanism for constructing functions of Hamiltonians. Our approach generalizes QSP by using $SU(2)$ rotations rather than X -rotations on the control ancilla similar to [20, 21]; however, our approach eschews the need to use $U^\dagger$ and avoids the use of Laurent polynomials in the analysis. Specifically, we use the convention that the signal operator is a 0-controlled application of $U = e^{iH}$:

$$A = (|0\rangle\langle 0| \otimes U) + (|1\rangle\langle 1| \otimes I) = \begin{bmatrix} U & 0 \\ 0 & I \end{bmatrix} \quad (5)$$

Next, instead of performing rotations in a single basis, we allow for signal processing operations to be arbitrary $SU(2)$ rotations of the ancillary qubit:

$$R(\theta, \phi, \lambda) = \begin{bmatrix} e^{i(\lambda+\phi)} \cos(\theta) & e^{i\phi} \sin(\theta) \\ e^{i\lambda} \sin(\theta) & -\cos(\theta) \end{bmatrix} \otimes I \quad (6)$$

Then by interleaving the above 2 operations as demonstrated in Figure 2, we can block encode polynomial transformations of the unitary matrix U without further assumptions. Note that below we have no assumptions made on the parity of the polynomial transformations unlike those made by the Theorem 1. Lastly, notice that we only need to specify a single λ since we can absorb the other instances into the instance of ϕ before them.

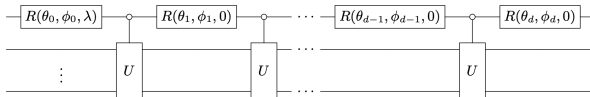


FIG. 2. Quantum circuit for GQSP wherein the angles θ and ϕ are chosen to enact a polynomial transformation of U . Here $R(\theta, \phi, \lambda)$ is a general single qubit rotation as specified in (6).

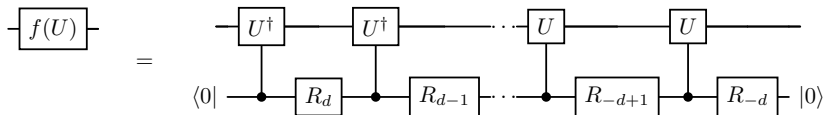
Theorem 3 (Generalized Quantum Signal Processing). $\forall d \in \mathbb{N}, \exists \vec{\theta}, \vec{\phi} \in \mathbb{R}^{d+1}, \lambda \in \mathbb{R}$ s.t.:

$$\left(\prod_{j=1}^d R(\theta_j, \phi_j, 0) A \right) R(\theta_0, \phi_0, \lambda) = \begin{bmatrix} P(U) \\ Q(U) \end{bmatrix} \quad (7)$$

$\longleftrightarrow$

1. $P, Q \in \mathbb{C}[x]$ and $\deg(P), \deg(Q) \leq d$.
2. $\forall x \in \mathbb{T}, |P(x)|^2 + |Q(x)|^2 = 1$.

Here $\mathbb{T} = \{x \in \mathbb{C} : |x| = 1\}$ is again the complex unit circle.



Pros

- Simple circuit using a single ancilla qubit.
- Any Laurent polynomial of degree d with $\|f\|_\infty \leq 1$ can be implemented exactly with $2d + 1$ uses of U .
- If f is analytic in a neighborhood of S^1 , then the error decreases exponentially in d .
- Constant scaling factor.

Cons (classical preprocessing)

- Need access to the coefficients β_j .
- Need to compute the rotation angles for $\{R_d, \dots, R_{-d}\}$ on a classical computer.

Pros

- Works for black-box function f . No need to access a (Laurent) polynomial approximation. The error scales with d as $E_d(f)$. If f is analytic in a neighborhood of S^1 , then the error decreases exponentially in d .
- Constant scaling factor.
- Only one call to U_f .
- Simple circuit and minimal classical preprocessing.

Cons

- $\mathcal{O}(\log(d))$ ancilla qubits.
- $8d - 2$ uses of U in comparison with $2d + 1$ uses required for QSP.

Let $f(z) = \sum_{j=-d}^d \beta_j z^j$ be a Laurent polynomial. Let us interpolate f by choosing $d_1 \geq 2d + 1$ equidistant points on S^1 .

Lemma

Let $f \in \mathbb{C}[z, z^{-1}]$ be a degree- d Laurent polynomial and $d_1 \geq 2d + 1$ an integer. Then for $\{z_k = \exp(2\pi i k / d_1), k = 0, \dots, d_1 - 1\}$,

$$f(z) = \frac{1}{d_1} \sum_{j,k=0}^{d_1-1} f(z_k) \left(\frac{z}{z_k} \right)^{j-d},$$

Proof follows from $\beta_j = (\sum_{k=0}^{d_1-1} f(z_k) z_k^{-j}) / d_1$.

Corollary

$$f(U) = \frac{1}{d_1} \sum_{j,k=0}^{d_1-1} f(z_k) \left(\frac{U}{z_k} \right)^{j-d},$$

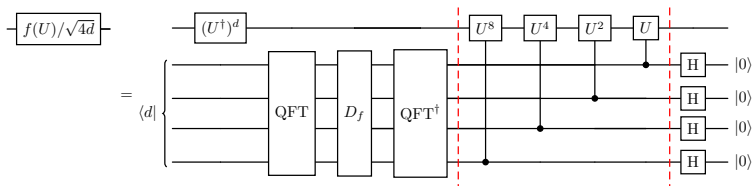
For simplicity, we will pick $d = 2^m$ and $d_1 = 4d$.

$$f(U) = \sum_{j=0}^{4d-1} \underbrace{\left(\frac{1}{4d} \sum_{k=0}^{4d-1} f(z_k) z_k^{d-j} \right)}_{\text{Fourier transform?}} U^{j-d} = \sum_{j=0}^{4d-1} \beta_{j-d} U^{j-d}.$$

So we can pick

$$\langle 0|U_L = \sum_{j=0}^{4d-1} \beta_{j-d} \langle j|, \quad \frac{1}{\sqrt{4d}} \sum_{j=0}^{4d-1} |j\rangle = U_R |0\rangle.$$

This leads to the circuit



The performance is slightly worse than the original LCU approach, but works for black-box f .

We are using $4d$ equidistant points and our Laurent polynomial is degree d , so we have $2d$ inequivalent formulae for interpolation!

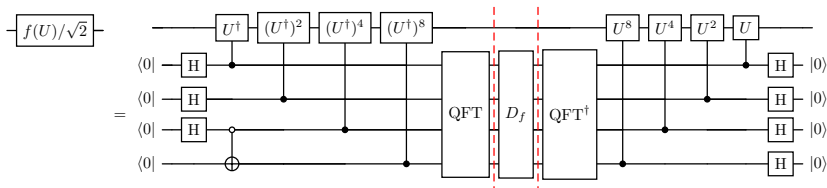
$$f(z) = \frac{1}{4d} \sum_{j,k=0}^{4d-1} f(z_k) \left(\frac{z}{z_k} \right)^{j-j'}, \quad j' \in \{d, d+1, \dots, 3d-1\}.$$

Now we can average to get

Lemma

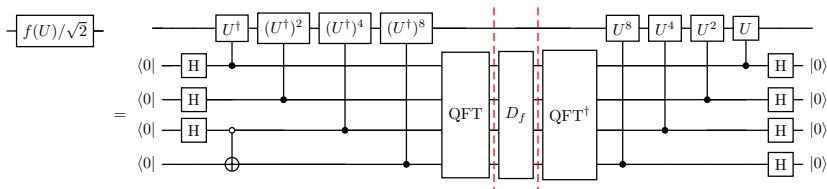
$$\begin{aligned} f(z) &= \frac{1}{8d^2} \sum_{j'=d}^{3d-1} \sum_{j,k=0}^{4d-1} f(z_k) \left(\frac{z}{z_k} \right)^{j-j'}, \\ \implies f(U) &= \frac{1}{8d^2} \sum_{j'=d}^{3d-1} \sum_{j,k=0}^{4d-1} f(z_k) \left(\frac{U}{z_k} \right)^{j-j'}. \end{aligned}$$

Constructing the block encoding



$$f(U) = \sqrt{2} \langle +_{2d} | \mathcal{W}_U^\dagger [\mathbb{1} \otimes \text{QPE} * D_f * \text{QPE}^\dagger] \mathcal{W}_U | +_{4d} \rangle.$$

Constructing the block encoding



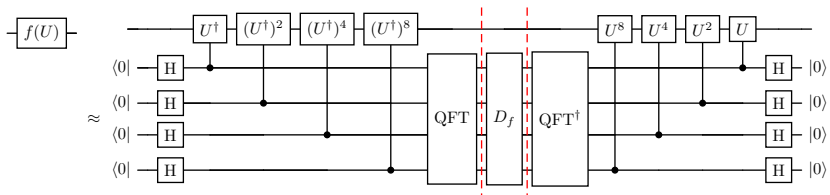
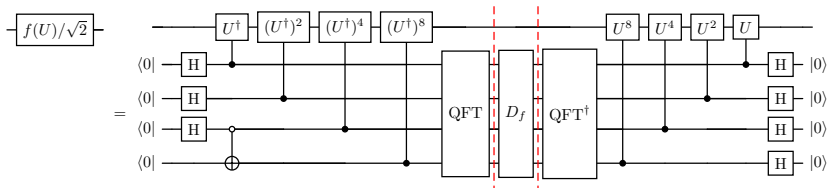
Pros

- A Laurent polynomial of degree d can be implemented exactly with $\mathbf{O}(d)$ uses of U (we will make a stronger claim soon!).
- Constant scaling factor.
- Only one call to U_f .
- Simple circuit and minimal classical preprocessing.

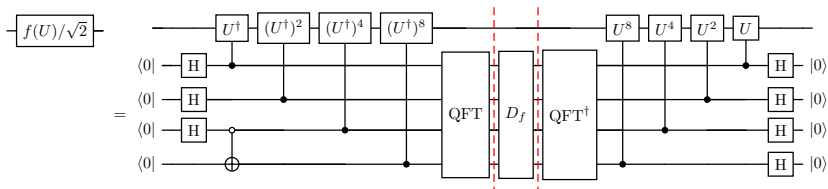
Cons

- $\mathbf{O}(\log(d))$ ancilla qubits.
- $8d - 2$ uses of U in comparison with $2d + 1$ uses in single-ancilla circuit.

Comparison with QPE-based approach



Comparison with QPE-based approach



Remark

The QPE block on the right computes phases to accuracy $\mathbf{O}(1/d)$, but the function implementation is exact as long as f is a degree- d Laurent polynomial.

Complexity of block encoding a function of a black-box unitary

Given an n -qubit black-box unitary U and a unitary $U_{f,4d}$ for $d = 2^m$, $m \in \mathbb{Z}^+$ encoding a function $f : S^1 \rightarrow \mathbb{C}$ with $\|f\|_\infty \leq 1$, we can construct a $(\sqrt{2}, m+3, (1+\sqrt{2})E_d(f))$ -block encoding of $f(U)$ by making $4d-1$ uses of $c-U$ and $c-U^\dagger$ each, one use of $U_{f,4d}$, and $\mathbf{O}(m^2)$ additional single- and two-qubit gates. Moreover, the circuit description for this block encoding can be computed using $\text{poly}(m)$ classical operations.

Reduction to function of a black-box unitary

Without loss of generality, we assume $U_H = U_H^\dagger$. Then we have the following reduction³

Lemma: Reduction to function of a unitary matrix

Let U_H be an $(\alpha, a, 0)$ -block encoding of an n -qubit Hamiltonian H , and let

$$W = (2\Pi - \mathbb{1})U_H, \quad \Pi = |0^a\rangle\langle 0^a|.$$

For any $g : [-1, 1] \rightarrow \mathbb{C}$,

$$g(H) = \langle 0^a | g\left((W + W^\dagger)/2\right) | 0^a \rangle.$$

Note

We can eliminate the negative powers of W using

$$\langle 0^a | W^j | 0^a \rangle = \langle 0^a | (W^\dagger)^j | 0^a \rangle = T_j(H),$$

where T_j is the j th Chebyshev polynomial of the first kind.

Reduced problem: Function of a black-box unitary

Given a function $f : S^1 \rightarrow \mathbb{C}$ with $\|f\|_\infty \leq 1$ and an n -qubit unitary U , construct a (α, a, ε) -block encoding of $f(U)$ with $\alpha \in \mathbf{O}(1)$.

Lemma

Let U_H be an $(n+a)$ -qubit self-inverse block encoding of an n -qubit Hamiltonian H , and let G be the Hermitian matrix with spectrum in $[-\pi, \pi)$ satisfying

$$W = (2\Pi - \mathbb{1})U_H, \quad \Pi = |0^a\rangle\langle 0^a|.$$

Then for any function $g : [-1, 1] \rightarrow \mathbb{C}$,

$$g(H) = \langle 0^a | g \left((W + W^\dagger)/2 \right) | 0^a \rangle.$$

It suffices to show that for any eigenstate $|\lambda\rangle$ of H , we have $\langle 0^a | g \left((W + W^\dagger)/2 \right) | 0^a \rangle |\lambda\rangle = g(\lambda) |\lambda\rangle$. To prove this equality, let

$$|\psi\rangle = |0^a\rangle |\lambda\rangle, \quad |\psi^\perp\rangle = (\mathbb{1} - \Pi)U_H |0^a\rangle |\lambda\rangle / \sqrt{1 - \lambda^2}.$$

Then

$$\begin{aligned} U_H |\psi\rangle &= \lambda |\psi\rangle + \sqrt{1 - \lambda^2} |\psi^\perp\rangle, \\ U_H |\psi^\perp\rangle &= \sqrt{1 - \lambda^2} |\psi\rangle - \lambda |\psi^\perp\rangle. \end{aligned}$$

Reduction to function of a black-box unitary

Using

$$(2\Pi - \mathbb{1})|\psi\rangle = |\psi\rangle, \quad (2\Pi - \mathbb{1})|\psi^\perp\rangle = -|\psi^\perp\rangle,$$

we get

$$\begin{aligned}(2\Pi - \mathbb{1})U_H|\psi\rangle &= \lambda|\psi\rangle - \sqrt{1-\lambda^2}|\psi^\perp\rangle, \\ (2\Pi - \mathbb{1})U_H|\psi^\perp\rangle &= \sqrt{1-\lambda^2}|\psi\rangle + \lambda|\psi^\perp\rangle.\end{aligned}$$

Therefore, the space $\mathcal{V} = \text{span}\{|\psi\rangle, |\psi^\perp\rangle\}$ is invariant under the action of $(2\Pi - \mathbb{1})U_H$, and furthermore, $(2\Pi - \mathbb{1})U_H$ acts as

$$[(2\Pi - \mathbb{1})U_H]_{\mathcal{V}} = \begin{bmatrix} \lambda & -\sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & \lambda \end{bmatrix}.$$

in this basis. Therefore

$$\begin{aligned}[W]_{\mathcal{V}} = \begin{bmatrix} \lambda & -\sqrt{1-\lambda^2} \\ \sqrt{1-\lambda^2} & \lambda \end{bmatrix} &\implies [(W + W^\dagger)/2]_{\mathcal{V}} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \\ &\implies [g((W + W^\dagger)/2)]_{\mathcal{V}} = \begin{bmatrix} g(\lambda) & 0 \\ 0 & g(\lambda) \end{bmatrix}\end{aligned}$$

as required.

Suppose U_H is not necessarily self-adjoint. Then

$$U'_H = |+\rangle\langle -| \otimes U_H + |- \rangle\langle +| \otimes U_H^\dagger$$

is a self-adjoint block encoding of H , where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$.

Proof: Easy to show that U'_H is unitary.

$$\langle 0| \langle 0^a| U'_H |0\rangle |0^a\rangle = \langle 0^a| (U_H + U_H^\dagger)/2 |0^a\rangle = H.$$

Note that U'_H can be constructed using one query each to U_H and $U_H^\dagger$.

Complexity of block encoding a function of a Hermitian matrix

Let $g : [-1, 1] \rightarrow \mathbb{C}$ with $|g(x)| \leq 1$ and let $f : S^1 \rightarrow \mathbb{C}$ be defined by $f(e^{i\theta}) = g(\cos \theta)$ for $\theta \in \mathbb{R}$. Given a $(1, a, 0)$ self-inverse block encoding U_H for an n -qubit Hamiltonian H and a unitary $U_{f, 4d}$ for $d = 2^m$, $m \in \mathbb{Z}^+$, we can construct a $(\sqrt{2}, a + m + 3, (1 + \sqrt{2})E_d(f))$ -block encoding $U_{g(H)}$ of $g(H)$ by making $8d - 2$ uses of $c-U_H$, one use of $U_{f, 4d}$, and $\mathbf{O}(da)$ additional two-qubit gates. Moreover, a circuit description of $U_{g(H)}$ can be computed using $\text{poly}(m)$ classical operations.

In QSVT, we are given a block encoding of an arbitrary matrix A with singular value decomposition $A = V_1^\dagger S V_2$ and a function g of definite parity, and the goal is to construct a block encoding of $g^{\text{SV}}(A)$, where $g^{\text{SV}}(A) = V_1^\dagger g(S) V_2$ if g is an odd function and $g^{\text{SV}}(A) = V_2^\dagger g(S) V_2$ if g is an even function. To perform QSVT, we first encode A in the off-diagonal block of a Hermitian matrix

$$H_A = \begin{bmatrix} 0 & A^\dagger \\ A & 0 \end{bmatrix}. \quad (1)$$

A block encoding of H_A can be constructed by making one use of the block encoding of A and its inverse respectively. Then we construct a block encoding of $g(H_A)$ to desired accuracy. We then invoke the fact that [?]

$$g^{\text{SV}}(A) = \begin{cases} \langle 0 | g(H_A) | 0 \rangle & \text{if } p = 0 \\ \langle 1 | g(H_A) | 0 \rangle & \text{if } p = 1 \end{cases}. \quad (2)$$

Lemma 40. (Polynomial approximations of $1/x$, [CKS17, Lemmas 17-19]) Let $\kappa > 1$ and $\varepsilon \in (0, \frac{1}{2})$. For $b = \lceil \kappa^2 \log(\kappa/\varepsilon) \rceil$ the odd function

$$f(x) := \frac{1 - (1 - x^2)^b}{x}$$

is ε -close to $1/x$ on the domain $[-1, 1] \setminus (-\frac{1}{\kappa}, \frac{1}{\kappa})$. Let $J := \lceil \sqrt{b \log(4b/\varepsilon)} \rceil$, then the $\mathcal{O}(\kappa \log(\frac{\kappa}{\varepsilon}))$ -degree odd real polynomial

$$g(x) := 4 \sum_{j=0}^J (-1)^j \left[\frac{\sum_{i=j+1}^b \binom{2b}{b+i}}{2^{2b}} \right] T_{2j+1}(x)$$

is ε -close to $f(x)$ on the interval $[-1, 1]$, moreover $|P(x)| \leq 4J = \mathcal{O}(\kappa \log(\frac{\kappa}{\varepsilon}))$ on this interval.

Lemma 11 (Block encoding inversion [37, Corollary 69]). *Let C be a matrix such that C/α_C is block encoded by O_C with some normalization factor $\alpha_C \geq \|C\|$. Then the operator*

$$\frac{C^{-1}}{2\alpha_{C^{-1}}} \tag{121}$$

can be block encoded with accuracy ϵ using

$$\mathbf{O}\left(\alpha_C \alpha_{C^{-1}} \log\left(\frac{1}{\epsilon}\right)\right) \tag{122}$$

queries to the controlled- O_C and its inverse, where $\alpha_{C^{-1}} \geq \|C^{-1}\|$ is an upper bound on norm of the inverse matrix.

Modulus of continuity

For a function $\tilde{f} : \mathbb{R} \rightarrow \mathbb{C}$ and a $\Delta \in \mathbb{R}^+$, the modulus of continuity $\omega(\Delta, \tilde{f})$ is defined to be

$$\omega(\Delta, \tilde{f}) = \sup_{|x_1 - x_2| \leq \Delta} \{ |\tilde{f}(x_1) - \tilde{f}(x_2)| \}.$$

Jackson's inequality

Let $f : S^1 \rightarrow \mathbb{C}$ be a function such that $\tilde{f}(\theta) = f(e^{i\theta})$ is r times differentiable. Then

$$E_d(f) \leq \frac{C(r)\omega(1/d, \tilde{f}^{(r)})}{d^r},$$

where

$$C(r) = \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^{k(r+1)}}{(2k+1)^{r+1}}$$

is a constant independent of f and d .

The constants $\{C(r)\}$ are known as Akhiezer-Krein-Favard constants. For $r = 0$ and $r = 1$, we have $C(0) = 1$ and $C(1) = \pi/2$ respectively.

Bound on the best Laurent polynomial approximation error

Let c be a non-integer positive constant, i.e. $c \in \mathbb{R}^+$, $c \notin \mathbb{Z}^+$. Let $g(x) = |x|^c$ and $f(e^{i\theta}) = g(\cos \theta) = |\cos \theta|^c$. Then

$$E_d(f) \in \mathbf{O}\left(\frac{1}{d^c}\right).$$

As a result, a block encoding of $|H|^c$ can be constructed using $\mathbf{O}(1/\varepsilon^{1/c})$ uses of $c\text{-}U_H$.

Remark on the case of analytic g

In general, one can achieve $\log(1/\varepsilon)$ complexity if $g(x)$ is analytic in a neighbourhood of $[-1, 1]$. Moreover, if g is analytic everywhere (i.e. entire), then $E_d(f)$ decreases faster than any exponential in d , so the query complexity is sublogarithmic in $1/\varepsilon$.